

SALES & REVENUE ANALYSIS DASHBOARD

Project Report

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Project Type: Data Analytics / Power BI Project

Domain: Sales and Business Analytics

Tool Used: Microsoft Power BI

1. Introduction

In today's business environment, organizations generate large amounts of sales data through customer transactions, products, regions, and different sales channels. Analyzing this data manually can make it difficult to understand business performance and identify important trends.

The **Sales & Revenue Analysis Dashboard** was developed using Microsoft Power BI to convert sales data into meaningful and interactive visualizations. The dashboard provides an overview of revenue, profit, orders, quantity sold, and profit margin while allowing users to analyze performance based on categories, regions, customer segments, products, and dates.

The main purpose of this project is to make sales information easier to understand and help users identify important business trends and performance areas.

2. Project Objective

The main objective of this project is to develop an interactive Power BI dashboard for analyzing sales and revenue performance.

The project focuses on the following objectives:

- Analyze overall sales and revenue performance.
 - Track important business KPIs such as revenue, profit, orders, quantity, and profit margin.
 - Identify the highest-performing product categories.
 - Identify the top 10 products based on revenue.
 - Analyze revenue performance across different regions.
 - Study monthly revenue trends.
 - Provide interactive filters for category, region, customer segment, and date.
 - Generate useful business insights from the available sales data.
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3. Dataset Description

The project uses a sales dataset containing information about customer transactions, products, sales performance, and revenue.

The major fields used in the analysis include:

Field	Description
Order_ID	Unique identification number for an order
Order_Date	Date on which the order was placed
Product	Name of the product
Category	Main product category
Sub_Category	Product sub-category
Customer_Segment	Type of customer
Region	Geographical sales region
City	Customer or sales location
State	State associated with the transaction
Quantity	Number of units sold
Unit_Price	Price per unit
Revenue	Revenue generated from the transaction
Cost	Cost associated with the transaction
Profit	Profit generated
Discount	Discount provided
Payment_Mode	Method of payment
Salesperson	Salesperson associated with the transaction

The dataset provides sufficient information to analyze sales, profitability, products, customers, and geographical performance.

4. Tools and Technologies Used

Microsoft Power BI

Power BI was used as the primary tool for data analysis and dashboard development. It was used to import the dataset, transform the data, create calculations, and build interactive visualizations.

Power Query

Power Query was used for preparing and transforming the data before analysis. It helps ensure that the data is in a suitable format for creating reports and visualizations.

DAX

DAX (Data Analysis Expressions) was used to create important calculated measures such as total revenue, total profit, total orders, total quantity, and profit margin.

Excel / CSV

The sales data was used as the source dataset for the analysis.

5. Data Preparation

Before creating the dashboard, the sales data was prepared for analysis.

The main data preparation activities included:

- Importing the sales dataset into Power BI.
- Checking the available columns and data types.
- Ensuring that numerical fields such as revenue, profit, quantity, and cost were correctly identified.
- Converting the order date into an appropriate date format.
- Checking the consistency of categorical fields such as category, region, and customer segment.
- Creating a separate date table for time-based analysis.
- Establishing relationships between the sales data and date table.
- Preparing the data for interactive filtering and visualization.

The prepared data was then used to create the dashboard.

6. DAX Measures

Several DAX measures were created to calculate the main performance indicators.

Total Revenue

Total Revenue = SUM(Sales_Data[Revenue])

This measure calculates the total revenue generated from all sales transactions.

Total Profit

Total Profit = SUM(Sales_Data[Profit])

This measure calculates the total profit generated from sales.

Total Quantity

Total Quantity = SUM(Sales_Data[Quantity])

This measure calculates the total number of products sold.

Total Orders

Total Orders = DISTINCTCOUNT(Sales_Data[Order_ID])

This measure counts the unique orders in the dataset.

Profit Margin

Profit Margin = DIVIDE([Total Profit], [Total Revenue], 0)

This measure calculates profit as a percentage of total revenue.

These measures were used throughout the dashboard to provide consistent and interactive calculations.

7. Dashboard Design

The dashboard was designed to provide a simple and interactive view of sales performance.

KPI Cards

Five KPI cards were created at the top of the dashboard:

- **Total Revenue:** 38.11M
- **Total Profit:** 10.22M
- **Total Orders:** 600
- **Total Quantity:** 2K
- **Profit Margin:** 26.82%

These KPIs provide a quick overview of the overall business performance.

8. Monthly Revenue Trend

A line chart was used to display the monthly revenue trend.

The visualization helps identify changes in revenue over time and makes it easier to observe periods of high and low sales performance.

The dashboard shows noticeable variation in monthly revenue, including a significant peak around **October 2025**.

This type of analysis can help businesses understand seasonal patterns and identify periods that require further investigation.

9. Revenue by Category

A bar chart was created to compare revenue across different product categories.

The dashboard shows that **Electronics** is the highest revenue-generating category, contributing approximately **28.4M** in revenue.

Other categories include:

- Furniture — approximately 6.0M
- Home Appliances — approximately 3.5M
- Office Supplies — approximately 0.2M

This indicates that Electronics is the primary contributor to overall revenue.

10. Top 10 Products by Revenue

A Top 10 Products chart was created to identify the products generating the highest revenue.

The visualization ranks products based on total revenue and helps identify the products that have the greatest contribution to sales performance.

This analysis can be useful for decisions related to:

- Inventory management
 - Product promotion
 - Marketing strategies
 - Product prioritization
 - Sales planning
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11. Revenue by Region

A donut chart was used to analyze revenue distribution across different regions.

The dashboard shows the following approximate revenue contribution:

- **South:** 13.5M
- **West:** 9.12M
- **North:** 7.82M
- **East:** 7.68M

The **South region** is the highest revenue-generating region in the dataset.

This regional analysis can help businesses identify stronger markets and evaluate opportunities for improving performance in lower-performing regions.

12. Interactive Filters and Slicers

One of the main features of the dashboard is its interactive filtering capability.

The dashboard includes slicers for:

Category

Allows users to analyze sales performance for a specific product category.

Region

Allows users to compare performance across different geographical regions.

Customer Segment

Allows users to analyze sales based on different customer groups such as Consumer, Corporate, and Small Business.

Date

Allows users to select a specific date range and analyze sales performance for that period.

When a filter is selected, the KPI cards and charts automatically update based on the selected data.

This makes the dashboard interactive and allows users to perform customized analysis.

13. Key Business Insights

The dashboard provides several useful insights from the sales data.

1. Electronics is the leading category

Electronics generates approximately **28.4M** in revenue, making it the strongest revenue-generating category.

2. South is the leading region

The South region contributes approximately **13.5M** in revenue and performs better than the other regions.

3. Overall revenue and profitability

The business generates approximately **38.11M in total revenue** and **10.22M in total profit**.

4. Healthy overall profit margin

The overall profit margin is approximately **26.82%**, indicating that a significant portion of revenue is retained as profit.

5. Revenue varies over time

The monthly revenue trend shows noticeable fluctuations, with a significant peak around October 2025. This suggests that there may be seasonal or time-based factors influencing sales.

6. Top products contribute significantly to revenue

The Top 10 Products analysis helps identify products that have a major impact on overall revenue and can be useful for future inventory and marketing decisions.

14. Business Value of the Dashboard

The dashboard can help businesses make data-driven decisions by providing a centralized view of sales performance.

For example, management can use the dashboard to:

- Monitor overall revenue and profit.
- Identify high-performing products.
- Identify the strongest sales categories.
- Compare regional performance.
- Understand customer segment performance.
- Analyze sales trends over time.
- Identify areas that may require additional marketing or sales efforts.

The interactive nature of the dashboard allows users to explore the data without manually creating separate reports for each category or region.

15. Challenges Faced

During the development of the project, some common challenges involved preparing the data correctly for analysis and creating meaningful visualizations.

One important part was ensuring that the date information was properly structured so that monthly revenue trends could be analyzed correctly.

Another challenge was creating appropriate DAX measures for calculating KPIs such as profit margin and total orders.

Designing the dashboard also required selecting visuals that communicated the information clearly without making the report too crowded.

16. Learning Outcomes

Through this project, I gained practical experience in:

- Importing and preparing data in Power BI.
- Understanding and cleaning sales datasets.
- Creating relationships between tables.
- Creating calculated measures using DAX.
- Building KPI cards.
- Creating interactive charts and slicers.
- Performing sales and revenue analysis.
- Identifying business insights from data.
- Designing a professional dashboard.
- Presenting analytical findings in a business-friendly format.

This project helped me understand how raw business data can be transformed into useful information for decision-making.

17. Conclusion

The **Sales & Revenue Analysis Dashboard** successfully provides an interactive view of sales performance using Power BI.

The dashboard combines key performance indicators, monthly trends, category analysis, regional analysis, top-performing products, and interactive filters into a single report.

The analysis shows that Electronics is the leading revenue category, South is the strongest region, and the overall business generates approximately 38.11M in revenue with a 26.82% profit margin.

Overall, the project demonstrates how Power BI, Power Query, and DAX can be used to transform raw sales data into meaningful business insights and support data-driven decision-making.

18. Future Enhancements

The dashboard can be further improved by adding:

- Sales forecasting for future months.
- Year-over-year growth analysis.
- Customer retention analysis.
- Profitability analysis at the product level.
- Drill-through pages for detailed product analysis.
- Geographic map visualizations.
- Automated data refresh.
- More advanced predictive analytics.

These additions could make the dashboard more useful for long-term business planning and decision-making.

19. Project Outcome

The final outcome of this project is an interactive **Sales & Revenue Analysis Dashboard** that allows users to monitor KPIs, analyze revenue trends, identify top products, compare categories and regions, and filter the results based on customer segment and date.

The project demonstrates practical knowledge of **data visualization, business analytics, Power BI, Power Query, and DAX**, making it suitable as a project-based internship and portfolio project.